Exact Query Obstructions in Characteristic-Two Modal Computation

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Provisional article skeleton: access contracts retained

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Abstract

This planned article studies exact all-outcomes-correct linear modal query procedures over characteristic-two fields. In a fixed pointwise oracle model, linear relations obstruct one-query Deutsch and Deutsch–Jozsa procedures, and an amplitude-polynomial span argument requires at least n queries for the n-bit Bernstein–Vazirani problem. The standard informative XOR oracle attains the Deutsch and Bernstein–Vazirani bounds with two and n queries. These upper bounds are not asserted for arbitrary coincident oracle gates. We distinguish genuine nilpotent-ring phase kickback from an invertible character readout, and retain the known UNIQUE-SAT protocol as a positive control against overbroad impossibility claims. The intended contribution is a precisely typed query and resource separation, subject to a remaining primary-source priority review, not a new polynomial method or a universal no-go for modal computation.

Drafting instruction. This is a planning skeleton, not a final paper. Read the exact contracts and correction C17 before inserting prose. Proof anchors are in `../../math/THEOREM_BANK.tex`, sections `sec:resources`, `sec:query` and `sec:diagnostics`.

1 Introduction

Drafting instruction. State Deutsch2, the one-query DJ obstruction and BVn with the standard XOR upper-bound qualifier immediately. Explain the difference between a structural lower bound and failure of a familiar circuit translation. Credit known modal UNIQUE-SAT as a counterweight, not a new discovery.

2 Oracle access and complete readout

Let F be a characteristic-two field. A pointwise oracle is $O_f = \bigoplus_x G_{f(x)}$ for fixed invertible target gates G_0, G_1. QXOR specializes this to $|x, y\rangle \mapsto |x, y + f(x)\rangle$. Preparations and nonquery linear operations are independent of the oracle. Ancillas are allowed. Every possible final outcome must carry the correct label; complete collected linear instruments must be injective.

Drafting instruction. Give exact dimensions/types, effect completeness and adaptive-record semantics. No postselection, unreported discarded branches or free access to full state descriptions is

included. Explain that arbitrary Boolean oracle logic can act as a linear basis permutation without being an affine Boolean function. Import that reversible-computation fact.

Canonical IDs: ALG-001.

Lemma 2.1 (Exact discrimination)**.** *For state sets $S_i \subset V \setminus \{0\}$, a complete injective linear readout distinguishes every label exactly if and only if the subspaces* $\operatorname{span} S_i$ *have a direct sum.*

Drafting instruction. Insert the coordinate-support necessity and basis-extension sufficiency. Compare modal distinguishability antecedents [10]; do not headline this elementary ingredient as a new discovery.

Figure specification: Exact-access diagram Include explicit adaptive branch records and a separate, differently labeled full-description-access model.

Figure 1: Planning placeholder; replace with the generated, source-linked figure.

3 One-query affine obstructions

Canonical IDs: ALG-002.

Theorem 3.1 (Deutsch)**.** *No exact one-query linear modal procedure under the fixed pointwise contract computes $f(0) + f(1)$. Under QXOR its exact query complexity is two.*

Drafting instruction. Prove $v_{00} + v_{11} = v_{01} + v_{10} = w$. Treat $w \neq 0$ by subspace intersection and $w = 0$ by identical states. Give the QXOR two-query basis-label construction. Explicitly state that $G_0 = G_1$ makes every oracle identical, invalidating any unqualified general upper bound.

Canonical IDs: ALG-003.

Theorem 3.2 (Deutsch–Jozsa)**.** *For $n \geq 2$ no exact one-query procedure satisfies the constant-or-balanced promise under the same contract. QXOR obeys $2 \leq Q_{\text{exact}} \leq 2^{n-1} + 1$.*

Drafting instruction. Use four equal address blocks and balanced supports AB, AC, BC whose three affine oracle operators sum to the constant-zero operator. Give the classical upper bound and explicitly leave its optimality in this model open. Do not infer a general multi-query theorem from the56-coordinate check.

4 Secret-polynomial span and Bernstein–Vazirani

Canonical IDs: ALG-004.

Theorem 4.1 (Exact BV)**.** *For $f_s(x) = s \cdot x$ with $s \in \mathbb{F}_2^n$, exact identification needs at least n queries under the fixed pointwise characteristic-two interface. QXOR attains n.*

Drafting instruction. Prove that O_s is affine in the secret and that after q queries all components are multilinear polynomials of degree at most q. All final states therefore span at most $\sum_{j\le q}\binom{n}{j}$ coefficient vectors, less than 2^n when $q < n$. Exact discrimination needs independence. Supply a full induction for complete recorded adaptive histories, including early stopping at different depths and maximum query count. Give the standard-address upper bound. Compare the established polynomial method [9] and explain why complex-amplitude BV does not use this same degree-one secret encoding.

5 Kickback is not a character basis

Canonical IDs: ALG-005, ALG-006, ALG-007.

Proposition 5.1 (Nilpotent-ring kickback and singular readout)**.** *In $\mathcal{A} = \mathbb{F}_2[u]/(u^4)$, let $z = 1 + u^2$. Then $X(1, z)^T = z(1, z)^T$, while $H_z = \begin{pmatrix} 1 & 1 \\ 1 & z \end{pmatrix} \sim \operatorname{diag}(1, u^2)$ is singular. The shared nth tensor power has binary rank $4 + 2n$.*

Drafting instruction. Insert the exact Smith/rank proof and the triviality of order-two multiplicative field characters in characteristic two. Explain that all A-unit characters reduce to one. Distinguish two-level R^2 kickback from a four-level R target, and shared from independent literal tensors. Do not claim all finite-field Fourier transforms or odd-order characters are obstructed.

Figure specification: Two different resources Show an actual kickback identity next to the rank-deficient character matrix, with the tensor base and dimensions written on each arrow.

Figure 2: Planning placeholder; replace with the generated, source-linked figure.

6 Positive control and scoped diagnostics

Canonical IDs: ALG-010, ALG-011, ALG-012; supporting ALG-008, ALG-009.

The Willcock–Sabry UNIQUE-SAT construction [6] remains a known one-query positive control with $O(n)$ other gates on its promise.

Drafting instruction. Report132 promised instances through $n = 6$, without relabeling the algorithm as ours. State that the finite Simon scan uses dimension16, 65,535 nonzero preparations,36 standard oracles,one query and no ancilla. Conventional Grover sign/diffusion failure is not an all-search impossibility. Present the full-description-readout ablation as a different model.

7 Related work and exact-model comparisons

Drafting instruction. Replace this source-navigation paragraph with a primary theorem-level comparison. Read [3, 4, 5, 7, 8] and query-related citations/dissertations. Record scalar algebra, oracle encoding, gate/readout sets, ancillas, description access and success notion. If named bounds

already exist under an equivalent or stronger access model, downgrade the novelty paragraph and reconsider article versus note format.

8 Limitations and open questions

No optimal unrestricted multi-query DJ bound, arbitrary-size Simon theorem, all-search no-go, runtime speed claim, probability distribution or physical realization follows. The field-based lower bounds and chosen ring kickback comparison are typed separately. A complete adaptive proof is required, not an appeal to a quantum deferred-measurement slogan. Priority of the exact bounds remains conditional until the remaining review is completed.

A Proof, experiment and source crosswalk

Drafting instruction. Place the full branch-record induction, upper-bound circuit details, phase-rank calculations and finite-control certificates here. Link all rerun commands and exact finite domains. Cite P01 or standard algebra rather than repeating a chain-ring resource-classification proof. Retain canonical IDs in an internal crosswalk even if journal theorem numbers change.

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